L10 Lab Packet — Worked Solutions

Wednesday, April 22, 2026 · Calculus II · University of Austin

How to use this document. Each part gives the full algebraic work followed by a short commentary on *why* the math says what it says, and *what* the answer means in context. The numerical results match the values you would have computed in class. If your computations differ, find the line where you diverged before reading further.

The solutions below are written in full prose; the lab packet itself uses fill-in blanks. Cross-reference by part letter (Warm-up, Part A, Part B, ...).

Problem 1: Dylan's Catalog

Setup. On December 7, 2020, Bob Dylan sold his songwriting catalog to Universal Music Publishing for \$300M. The catalog generates roughly $R_0 = \$10.7\text{M/year}$ in royalties. Take a 20-year horizon and a 4% discount rate. Question: who got the better deal?

Warm-up (a) — Discount one payment.

A one-time payment of \$100 is due at year 5 with $r = 4\%$. What is it worth today?

A dollar in an account earning continuous rate r grows from X today to Xe^{rt} at time t. Reading this backward: a dollar received at time t is worth e^{-rt} today. So

$$\text{PV} \;=\; 100 \cdot e^{-0.04 \cdot 5} \;=\; 100\, e^{-0.2} \;\approx\; \$81.87.$$

General formula: $\text{PV} = C\, e^{-rt}$.

Warm-up (b) — Forward to year 20.

The payee leaves that \$100 in an account earning $r = 4\%$ until year 20. What is it worth at year 20?

The \$100 arrives at year 5 and grows for $T - t = 20 - 5 = 15$ more years:

$$\text{FV} \;=\; 100 \cdot e^{0.04 \cdot 15} \;=\; 100\, e^{0.6} \;\approx\; \$182.21.$$

General formula: $\text{FV} = C\, e^{r(T-t)}$.

The exponent $T - t$, not T. The money only earns interest for the time *after* it arrives. A payment at $t = 0$ would grow for the full T years; a payment at $t = T$ doesn't grow at all. The exponent measures elapsed time from arrival to the valuation date.

Part A — What does the PV integral say?

In one sentence, what question does $\text{PV} = \int_0^T R_0\, e^{-rt}\, dt$ *answer? How does the integrand relate to the warm-up formula?*

The integral answers: *how much money do I need today, in an account paying r, to replicate this entire stream of royalties?*

Part B — Is the FV the same integral?

If Dylan had kept the catalog and reinvested each year's royalties at rate r, what would he have at year 20?

No, it is not the same integral. PV walks each slice *back* to time 0 (factor e^{-rt}). FV walks each slice *forward* to time T (factor $e^{r(T-t)}$).

Part C — Set up and evaluate the FV integral.

A slice $R_0\,dt$ earned at time t grows for $(T-t)$ years to become $R_0\,e^{r(T-t)}\,dt$ at year T (Warm-up (b) per slice). Sum:

$$\text{FV} \;=\; \int_0^T R_0\,e^{r(T-t)}\,dt.$$

Evaluation by u-substitution. Let $u = T - t$, so $du = -dt$. When $t = 0$, $u = T$; when $t = T$, $u = 0$:

$$\text{FV} \;=\; \int_T^0 R_0\,e^{ru}\,(-du) \;=\; \int_0^T R_0\,e^{ru}\,du \;=\; R_0 \cdot \frac{e^{rT}-1}{r}.$$

Faster route. Pull e^{rT} out of the integrand:

$$\text{FV} \;=\; e^{rT}\int_0^T R_0\,e^{-rt}\,dt \;=\; e^{rT}\cdot\text{PV}.$$

Both routes give the same closed form: $\text{FV} = R_0\big(e^{rT}-1\big)/r = \text{PV}\cdot e^{rT}$.

The relation $\text{FV} = \text{PV}\cdot e^{rT}$. Either viewpoint values the same stream of cash. If you have the present value, growing it forward T years at rate r must give the future value. So the two integrals are not independent answers — they are the same number expressed at two different times.

Part D — Compute the numbers.

Use $R_0 = 10.7$, $T = 20$, $r = 0.04$.

$$e^{rT} \;=\; e^{0.04\cdot 20} \;=\; e^{0.8} \;\approx\; 2.2255.$$

$$\text{PV} \;=\; \frac{10.7\,(1-e^{-0.8})}{0.04} \;=\; \frac{10.7\cdot(1-0.4493)}{0.04} \;\approx\; \frac{10.7\cdot 0.5507}{0.04} \;\approx\; \$147.3\text{M}.$$

$$\text{FV} \;=\; \frac{10.7\,(e^{0.8}-1)}{0.04} \;=\; \frac{10.7\cdot 1.2255}{0.04} \;\approx\; \$327.8\text{M}.$$

Sanity check: $\text{PV}\cdot e^{rT} = 147.3\cdot 2.2255 \approx 327.8$. ✓

The \$300M Dylan got, valued at year 20:

$$300 \cdot e^{0.8} \approx 300 \cdot 2.2255 \approx \$667.7\text{M}.$$

Who won the deal (under these assumptions)? Compare in either viewpoint:

- *Today (PV).* Dylan's \$300M cash sits as-is. The 20-yr royalty stream is worth PV = \$147.3M today. \$300M > \$147M.
- *Year 20 (FV).* Grow Dylan's cash forward at the same rate $r = 4\%$:

$$300 \cdot e^{rT} = 300 \cdot e^{0.8} \approx 300 \cdot 2.2255 \approx \$667.7\text{M}.$$

  The 20-yr royalty stream is worth FV = \$327.8M at year 20. \$668M > \$328M.

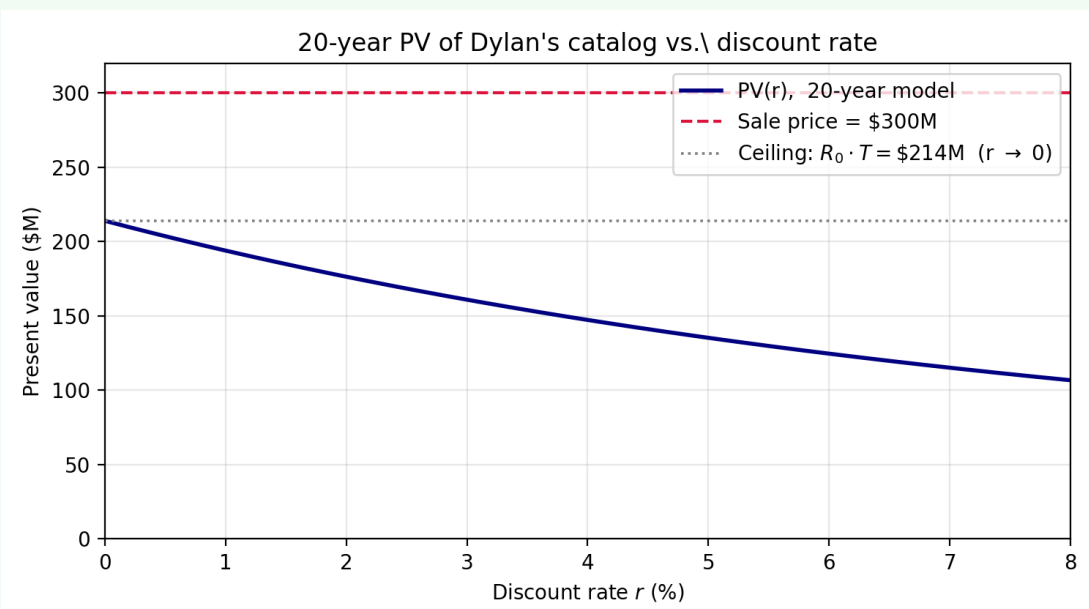
(The two viewpoints differ by exactly a factor of $e^{rT} \approx 2.226$ on each side, as the previous box predicts.)

Either way, the gap is the same fact: **Dylan came out ahead by roughly a factor of two**, under a 20-yr horizon at $r = 4\%$. Part F will revisit whether the model itself is the problem.

Part E — Sweep the discount rate.

Plot $\text{PV}(r) = R_0(1 - e^{-rT})/r$ *for* r *from 0 to 8%. Mark \$300M. Where does the curve cross?*

It never crosses \$300M. See the plot below (generated in `L10_Catalog.ipynb`, Part E cell):



The navy curve is $\text{PV}(r)$. It is highest at $r = 0$ and decreases as r grows (a higher discount rate makes future dollars worth less). The crimson dashed line is the \$300M sale price. **The crimson line sits above the entire navy curve**, so the equation $\text{PV}(r) = 300$ has no solution for any $r \geq 0$.

The ceiling at $r = 0$. If we don't discount at all, every $\$R_0\,dt$ slice of royalties keeps its full

value, so the PV is just the undiscounted total:

$$\mathrm{PV}(0) \;=\; \int_0^T R_0\,dt \;=\; R_0 \cdot T \;=\; 10.7 \cdot 20 \;=\; \$214\mathrm{M}.$$

That \$214M is the gray dotted ceiling on the plot. Any positive r gives a smaller PV.

Why the closed form looks like $0/0$ at $r = 0$. Plugging $r = 0$ into $R_0(1 - e^{-rT})/r$ gives $0/0$, but the underlying integral $\int_0^T R_0\,e^{-rt}\,dt$ is perfectly well defined at $r = 0$: the integrand becomes the constant R_0, and the area is $R_0 \cdot T$. The closed form has a removable singularity there; the integral does not. That is: the formula needs patching at $r = 0$ (there's a single point where the symbols don't evaluate, and you fix it by filling in the limit R_0T), but the integral needs no patching at all — at $r = 0$ the integrand is just the constant R_0, so the integral evaluates straight to R_0T like any other definite integral.

Part F — Read the gap.

The 20-year model says the stream is worth at most \$214M. Universal paid \$300M. What does the gap tell us?

The gap is \$300M − \$214M = \$86M, which the 20-yr model cannot justify at any discount rate. Something in the model has to give. Three reasonable suspects:

1. **Horizon too short.** Royalties don't stop at year 20. A copyright on Dylan's songs runs for the author's life plus 70 years; in practice the catalog will pay out for many decades.
2. **Growth missing.** We assumed R_0 flat. Streaming revenue has grown sharply, and a catalog of canonical songs may be a growing income stream rather than a flat one.
3. **Universal overpaid.** Always on the table.

Part G explores suspect (1) by extending the horizon to infinity.

Part G — The perpetuity model.

Set up the perpetuity integral. Set $\mathrm{PV}_\infty = 300$ *and solve for the implied rate.*

The perpetuity PV is the same integrand with $T \to \infty$:

$$\mathrm{PV}_\infty \;=\; \int_0^\infty R_0\,e^{-rt}\,dt \;=\; R_0\Big[-\tfrac{1}{r}e^{-rt}\Big]_0^\infty \;=\; \frac{R_0}{r}.$$

This improper integral is finite because the antiderivative $-e^{-rt}/r$ goes to 0 as $t \to \infty$ (since $e^{-rt} \to 0$), so the upper limit contributes nothing and only the value at $t = 0$ survives. Equivalently: the leftover area beyond any cutoff T is $(R_0/r)\,e^{-rT}$, which shrinks to 0 as $T \to \infty$.

Setting $\mathrm{PV}_\infty = 300$:

$$\frac{R_0}{r} = 300 \quad \Longrightarrow \quad r = \frac{R_0}{300} = \frac{10.7}{300} \approx 0.0357 = 3.57\%.$$

Reading the implied rate. Think of Universal's \$300M like a loan: they paid \$300M upfront, and the catalog "pays them back" \$10.7M/year forever. The implied r is the annual return they earn on that \$300M.

Compared to long-run benchmarks:

Asset	Long-run nominal return
S&P 500 (since 1957)	∼10%/yr
Investment-grade corp. bonds	∼4%/yr
10-yr Treasuries	∼3%/yr
Bank savings	∼0%/yr
Universal's implied rate	**3.57%/yr**

3.57% sits below the corporate-bond yield and far below the stock-market average. So Universal is implicitly betting the catalog will (a) outlast a bond's typical maturity *and/or* (b) grow over time — streaming revenue trends up — enough to compensate for accepting a lower upfront yield than safer alternatives. "Universal overpaid" is a possible reading; "Universal made a bet on catalog durability and growth" is the more interesting one.

Pivot: From Finance to Medicine

The same three tools (exponentials, u-substitution, geometric series) reappear in Problem 2 with new labels:

Tool	**Finance (P1)**	**Pharmacology (P2)**
Exponential decay	e^{-rt} (discount)	e^{-kt} (elimination)
u-sub integral	$R_0(1-e^{-rT})/r$	$\mathrm{AUC} = D/(V_d k)$
Geometric series	$\mathrm{PV}_\infty = R_0/r$	steady-state trough

What changes is the units, the interpretation, and the stakes.

Problem 2: Vancomycin Dosing

Setup. A 70-kg ICU patient with MRSA sepsis. Design a dosing schedule for vancomycin so the steady-state trough lands in the therapeutic window 15–25 mg/L. After a single IV dose D, the concentration follows $C(t) = C_0\, e^{-kt}$ with $C_0 = D/V_d$. Use $V_d = 50\,\mathrm{L}$ and $k = 0.10/\mathrm{hr}$.

Part A — Single-dose AUC.

Compute $\mathrm{AUC} = \int_0^\infty C_0\, e^{-kt}\, dt$*, then plug in* $D = 1000\,mg$*.*

Apply FTC with antiderivative $-\frac{1}{k}e^{-kt}$:

$$\text{AUC} \;=\; \int_0^\infty C_0\, e^{-kt}\, dt \;=\; C_0\Big[-\tfrac{1}{k}e^{-kt}\Big]_0^\infty \;=\; C_0 \cdot \tfrac{1}{k} \;=\; \frac{C_0}{k}.$$

The improper integral is finite because the antiderivative goes to 0 as $t \to \infty$ (since $e^{-kt} \to 0$), so the upper limit contributes nothing and only the value at $t = 0$ survives. (Same convergence argument as the perpetuity in Part G of Problem 1; the integrals have identical form.) Substituting $C_0 = D/V_d$:

$$\text{AUC} \;=\; \frac{D}{V_d\, k}.$$

With $D = 1000\,\text{mg}$, $V_d = 50\,\text{L}$, $k = 0.10/\text{hr}$:

$$\text{AUC} \;=\; \frac{1000}{50 \cdot 0.10} \;=\; \frac{1000}{5} \;=\; 200\ \text{mg} \cdot \text{hr/L}.$$

What AUC means. Concentration (mg/L) integrated over time (hr) gives a "drug-hour" quantity — the total exposure of the body to the drug from one dose. Same algebraic structure as "speed × time = distance." Clinically, vancomycin efficacy tracks the 24-hr AUC, with a target around 400–600 mg·hr/L. A single 1000 mg dose delivers 200, well short of the daily target: we will need to dose repeatedly.

Structural parallel to Problem 1. The perpetuity integral $\text{PV}_\infty = R_0/r$ has the exact same form: a constant times an exponential decay, integrated to infinity, gives constant/rate. Replace "money now / discount rate" with "concentration now / elimination rate."

Part B — Repeated dosing and the steady-state trough.

Give dose D every τ hours. Write the trough concentration just before the $(n{+}1)$-th dose as a sum, collapse to closed form, take $n \to \infty$.

Identify the missing term. Just before the $(n{+}1)$-th dose, dose j was given $(n - j + 1)\tau$ hours ago, so its leftover concentration is $C_0\, e^{-k(n-j+1)\tau}$. In particular, the n-th dose (given τ hr ago) contributes $C_0\, e^{-k\tau}$.

Sum over all prior doses. Reindex with $i = n - j + 1$ (hours-ago counter, ranging $i = 1, 2, \ldots, n$):

$$C_{\text{trough}}(n) \;=\; \sum_{i=1}^{n} C_0\, e^{-ik\tau} \;=\; C_0 \sum_{i=1}^{n} \big(e^{-k\tau}\big)^i.$$

This is a finite geometric series with first term $a = C_0\, e^{-k\tau}$ and common ratio $\rho = e^{-k\tau}$. Closed form:

$$C_{\text{trough}}(n) \;=\; C_0\, e^{-k\tau} \cdot \frac{1 - e^{-nk\tau}}{1 - e^{-k\tau}}.$$

Steady state. As $n \to \infty$, the factor $e^{-nk\tau} \to 0$, so the numerator $\to 1$:

$$\boxed{C_{\text{ss, trough}} = \frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}}\,.}$$

Why this converges. Each new dose adds a smaller contribution to the trough than the previous one: dose n contributes $C_0 e^{-k\tau}$, dose $n-1$ contributes $C_0 e^{-2k\tau}$, etc. Because the ratio $e^{-k\tau}$ is strictly less than 1, the sum converges. The body's clearance and the dosing schedule reach an equilibrium: each cycle, the patient eliminates exactly as much drug as one dose delivers.

Same structure as a perpetuity. The L09 discrete perpetuity $\sum_{n=1}^{\infty} R\,(1+r)^{-n} = R/r$ and the trough sum here are the same kind of object: a constant times a geometric ratio less than 1, summed from $n = 1$ to ∞, gives a finite total. Different units, identical algebra.

Part C — Try $D = 1000$ mg, $\tau = 12$ hr (Q12h).

$$C_0 = \frac{D}{V_d} = \frac{1000}{50} = 20 \text{ mg/L}.$$

$$e^{-k\tau} = e^{-0.10\cdot 12} = e^{-1.2} \approx 0.3012.$$

$$C_{\text{ss, trough}} = \frac{C_0\, e^{-k\tau}}{1 - e^{-k\tau}} = \frac{20 \cdot 0.3012}{1 - 0.3012} = \frac{6.024}{0.6988} \approx 8.62 \text{ mg/L}.$$

Does this land in the 15–25 mg/L window? If not, which way is it off, and what would you change?

8.62 mg/L is below the 15 mg/L MEC. The infection would get a growth window each cycle. Two levers to fix it:

1. **Shorten** τ so less drug clears between doses (e.g., go from Q12h to Q8h).
2. **Raise** D so each dose contributes more.

Clinically, lever (1) is preferred — it raises the trough without raising the peak as much.

Try Q8h with the same dose ($D = 1000$ mg, $\tau = 8$ hr):

$$e^{-k\tau} = e^{-0.8} \approx 0.4493, \qquad C_{\text{ss, trough}} = \frac{20 \cdot 0.4493}{1 - 0.4493} = \frac{8.986}{0.5507} \approx 16.3 \text{ mg/L}.$$

This lands inside the 15–25 mg/L window, so Q8h at 1000 mg is a viable regimen.

Why shorter intervals raise the trough. The trough is the leftover from *all* prior doses, weighted by how long each has been clearing. Shortening τ does two things: (a) each individual dose has had less time to decay before the next arrives, so the leading term $C_0\, e^{-k\tau}$ is larger, and (b) the geometric sum has a larger common ratio $e^{-k\tau}$, so more historical doses contribute meaningfully. Both push the trough up.

Part D — Concentration profiles and a missed dose (Colab).

Plot $C(t)$ for Q12h and Q8h regimens over 5 days. Find when the profile stabilizes. Re-run Q8h skipping dose #6.

(1) Profiles. The Q12h sawtooth oscillates with troughs $\approx 8.6\,\text{mg/L}$ (below the $15\,\text{mg/L}$ MEC line); peaks reach $\approx 28.6\,\text{mg/L}$ (above MTC briefly). The Q8h sawtooth settles into the 15–25 window with troughs ≈ 16.3 and peaks $\approx 36.3\,\text{mg/L}$.

(2) When steady state arrives. Vancomycin half-life $t_{1/2} = \ln 2/k \approx 6.93\,\text{hr}$. Roughly 4–5 half-lives are needed to be within 5% of steady state, so the locked-in regime arrives by about 28–35 hr (4–5 doses on Q8h). After that, peaks and troughs are visually indistinguishable from cycle to cycle.

Definition reinforced. Steady state is not "in a band." The first few doses are in a band too — a band that's still climbing. Steady state is when *the band stops moving*: each cycle's peak equals the previous cycle's peak, and likewise for troughs.

(3) Missed dose #6 on Q8h. The concentration drops below MEC for roughly 4–6 hr before the next (#7) dose pulls it back up. The patient is briefly exposed to subtherapeutic levels but recovers within one dose interval; no toxicity excursion.

Does *when* you skip matter? A miss during ramp-up (e.g., dose 3) leaves a longer below-MEC tail because there is less accumulated drug to fall back on. A miss at steady state (dose 9 or later) recovers fastest because the accumulated reservoir cushions the dip.

Part E — Interpret the missed dose.

Is one missed dose of Q8h vancomycin a medical emergency, a minor setback, or somewhere in between?

Minor setback. The plot supports this read directly:

- Concentration dips below MEC for ∼4–6 hr (about half a dose cycle).
- No toxicity excursion — nothing crosses the MTC.
- The next scheduled dose pulls concentration back into the window.

"Emergency" overstates it (the breach is short, no toxicity, recovery is automatic). "Nothing happened" understates it (the bacteria did get a sub-MEC growth window, however brief). The honest read is in between: a real but bounded gap, recovered by the next dose, with a slight context-dependence on *when* the miss occurred.

What this says about regimen design. Because steady state acts as a reservoir, single missed doses during steady state are recoverable. Missed doses during ramp-up are riskier — a reason loading doses (a larger first dose) are sometimes used to reach therapeutic levels faster and shorten the vulnerable window.

Wrap-up

Same tools, two worlds. Block 2 gave you three machines:

- *Exponential decay* as an integrand: e^{-rt} for money, e^{-kt} for drug.
- *u-substitution* on $\int e^{at}\,dt$ to evaluate the resulting closed form.
- *Geometric series* to sum infinitely many decayed contributions.

You used each of them twice today, in problems whose surface stories look nothing alike. The integral $\int_0^\infty R_0\,e^{-rt}\,dt = R_0/r$ and the integral $\int_0^\infty C_0\,e^{-kt}\,dt = C_0/k$ are the same calculation. The geometric trough sum $C_0 \sum e^{-jk\tau}$ and the perpetuity sum $\sum (1+r)^{-n} R$ are the same calculation.

The math does not care about the units.

Memo deliverable

Your personalized handout (one of six artist catalog sales) gives you the full prompt. In brief:

1. Fill in the four-row table: 20-yr PV, max possible PV (i.e. R_0T), sale price, implied perpetuity rate.
2. Write a memo (≤ 1 page) explaining whether the 20-yr model can justify the price, and what the implied perpetuity rate says about the buyer's assumptions.

Due Friday, April 24, with HW2.